

# TAMILSELVAN G

**Software Engineer | .NET Full Stack Developer | Angular | ASP.NET Core | AI / RAG**

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## PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building and maintaining enterprise web applications using Angular, TypeScript, C#, ASP.NET Core, SQL Server and MongoDB. Experienced in developing features across frontend, backend and database layers, with hands-on ownership of application modules from initial setup through deployment and production support. Worked on publishing, workforce management, project management and email-processing applications, including AI-assisted features for email classification and summarization. Also built a personal Python-based RAG application using FastAPI, Qdrant, Redis, Google Gemini and Ollama.

## TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL, Python

Frontend: Angular 14–20, RxJS, HTML5, CSS3, Tailwind CSS, Bootstrap, Angular Material, Reactive Forms, Lazy Loading, ApexCharts, TinyMCE

Backend: ASP.NET Core, C#, Web API, REST APIs, SignalR, Entity Framework Core, Dependency Injection, Middleware, JWT Authentication, RBAC, Clean Architecture

Databases: SQL Server, MongoDB, Redis

Testing & API Tools: Manual Testing, API Testing, Postman, Swagger/OpenAPI, Chrome DevTools, Debugging

Cloud, DevOps & Hosting: Docker, Docker Compose, Git, GitHub Actions, AWS S3, IIS

AI & Integration: RAG, LLM Integration, Google Gemini API, Ollama, Vector Embeddings, Qdrant, SSE Streaming, Prompt Engineering

Development Practices: Agile/Scrum, Code Review, Troubleshooting, Performance Optimization

## PROFESSIONAL EXPERIENCE

### Nova Techset Pvt. Ltd. - Chennai, India

Associate Software Engineer | Sep 2022 - Present

Junior Software Engineer | Mar 2023 - Aug 2024

Trainee Software Engineer | Jun 2022 - Feb 2023

- Progressed from Trainee Software Engineer to Associate Software Engineer while working on multiple enterprise applications for publishing and business operations.
- Developed full-stack features using Angular, TypeScript, ASP.NET Core, C#, SQL Server and MongoDB, working across UI, REST APIs, database operations and application integration.
- Worked on five enterprise applications across publishing workflows, content editing, workforce management, project management and employee profiling.
- Improved backend efficiency through SQL query optimization, indexing, Redis caching and application-level improvements; project-level improvements were estimated at approximately 15%.
- Improved Angular application performance using lazy loading, RxJS optimization, component-level improvements and frontend profiling.
- Designed and implemented REST APIs with ASP.NET Core and C#, including authentication, authorization and role-based access control for enterprise applications.
- Implemented real-time application functionality using SignalR for collaborative publishing and workflow-related features.
- Worked with MongoDB for project management and email-processing workflows, including conversation data, message tracking and dashboard-related queries.
- Worked with AWS S3-based file upload and download functionality and reused the established file-handling approach across applications.
- Containerized applications using Docker and Docker Compose and supported application hosting and deployment through IIS.
- Performed manual testing, API testing, debugging and issue investigation using Postman, Swagger/OpenAPI and Chrome DevTools.
- Collaborated with development and AI teams to integrate AI-generated email summaries and classification results into enterprise dashboards.

## KEY PROJECTS

### 1. nPMS - Project Management & Email Monitoring System

Technologies: Angular 20, TypeScript, .NET Core 10, C#, MongoDB, REST API, JWT, AI Integration

- Own end-to-end frontend and backend development for an enterprise project management and email monitoring system used to track project-related PM emails.
- Built email monitoring and classification workflows that categorize incoming emails based on business requirements and present the results through role-based dashboards.

- Developed hierarchical dashboards for management, publishers, managers, team leads and project managers, providing visibility into pending emails and email-related activities.
- Implemented tracking for email response and SLA-related metrics, including pending responses, responses after 20 hours and other operational status information.
- Integrated AI-generated email summaries and classification/reason information into the application to help users understand email context more quickly.
- Worked with the AI team to consume processed email data and display AI-generated information within the application.
- Implemented MongoDB-based conversation and message handling for email threads, including received/sent message information and conversation-level status tracking.

## 2. NVMS - Vendor Management System

Technologies: Angular, TypeScript, Tailwind CSS, .NET Core, C#, SQL Server, Redis, Docker, JWT, REST API

- Developed and maintained an enterprise vendor management platform supporting publishing workforce operations and managing a network of 300+ freelancers.
- Built frontend modules for vendor onboarding, qualification, service allocation, invoice management and operational workflows.
- Implemented role-based access and secured application functionality using JWT-based authentication and authorization.
- Developed vendor competency and assessment workflows to capture publishing-specific skills and qualification results.
- Worked on invoice and billing workflows involving word counts, deliverable types, rates and payment-related information.
- Used Redis caching for frequently accessed application data and optimized backend/database operations for commonly used APIs.
- Worked with existing AWS S3 file upload and download functionality and reused the established implementation for application requirements.

## 3. NEDIT - Web-Based Content Editing & XML Proofing Platform

Technologies: Angular, TypeScript, TinyMCE, ICE Plugin, SignalR, .NET Core, C#, SQL Server, XML

- Contributed to an enterprise web application used for content editing, XML validation and proofing workflows in publishing operations.
- Developed rich-text editing functionality using TinyMCE, including formatting, paragraph styling, footnotes and image-related content operations.
- Implemented and enhanced change-tracking functionality using the ICE Plugin for revision history and content comparison.
- Worked on SignalR-based real-time collaboration features to synchronize editing and workflow updates between users.
- Developed features for mathematical content editing and integrated spell-checking and XML validation into the publishing workflow.
- Worked on maintaining XML structure and content integrity throughout editing and proofing processes.

## 4. Sesame - Article Tracking & Publishing Workflow System

Technologies: Angular, TypeScript, ASP.NET Core, C#, SQL Server, REST APIs

- Worked on an enterprise publishing application used to track articles and manage their workflow through different stages.
- Developed and maintained multiple master pages used to configure and manage application-level business data.
- Worked across frontend and backend layers to implement business requirements and connect Angular screens with ASP.NET Core APIs. Handled UI development, API integration, form validation and data management for assigned modules.

## PERSONAL PROJECT

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### AI RAG Assistant - Document Question-Answering Application

Technologies: Python, FastAPI, Qdrant, Redis, Google Gemini API, Ollama, Docker, GitHub Actions, Render

- Built a personal Retrieval-Augmented Generation (RAG) application that allows users to upload PDF, DOCX, TXT and CSV documents and ask questions against their content.
- Implemented document ingestion, text extraction, chunking, embedding generation and vector search using Python, Qdrant and Google Gemini.
- Used Qdrant for vector storage and similarity search to retrieve relevant document chunks before generating answers.
- Integrated Google Gemini for LLM-based responses and used Ollama with local models such as bge-m3 and qwen2.5:7b during local development and experimentation.
- Used Redis for conversation memory and application-level caching.
- Built REST APIs with FastAPI and organized the application using dependency-based services to keep the components easier to maintain and test.
- Containerized the application using Docker Compose and created a GitHub Actions CI/CD workflow for deployment to Render.
- Used AI coding tools extensively during development for implementation assistance, debugging and development support while learning and working through the application's architecture and components.

## EDUCATION

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Bachelor of Engineering - Computer Science and Engineering

Alpha Engineering College, Anna University | 2021 | CGPA: 7.45 / 10